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SUMMARY

I do not just fix the security gaps I am told about. I try to understand a system deeply enough that I can spot the ones nobody has flagged yet, which is why the access control and anti fraud logic I have built comes from asking how the system could fail, not from ticking off a checklist. Before I write a line of code, I am already tracing how data moves through the whole application, so the database schema ends up being a map of how the system actually works, not just a structure to fill in. I am a final year Diploma in ICT Application Development student at DUT, looking for .NET Developer, Software Developer, or Full Stack Development roles where I can keep building software this way.

PROFESSIONAL EXPERIENCE

Geeked-On x Tech Horizons IT Incubation Program

June 2026 –

July 2026

Durban , KZN

- Database design was not just about creating tables, it was about mapping how data would need to move before writing any code. I architected relational database structures in SQL Server and used Entity Framework Core to configure database migrations and manage primary and foreign key navigation properties, so the schema reflected how the application actually needed to work.
- Actively developing full stack web applications utilizing the ASP.NET MVC framework and C#, ensuring strict adherence to architectural patterns and clean separation of concerns.
- Working within a sprint style team environment of six developers to build enterprise software from scratch for a Demo Day milestone, actively utilizing Git for version control and GitHub for repository management and team code merging.
- Actively used AI tools and prompt engineering methodologies to accelerate frontend UI construction and optimize backend business logic pipelines, reducing development time across a 6 member sprint team building an enterprise grade web application from scratch for a public Demo Day milestone.

KZN Imbokodo Hackathon 2026 | Participant

- Conducted user centric research on menstrual equity and female entrepreneurship to define problem statements, buyer and seller personas, and B2B business models for local micro enterprises. This research became the foundation for the system design work now detailed under the HerCycle project.
- Participated in an intensive 24 plus hour coding sprint, collaborating closely with teammates and industry mentors to deliver a full product pitch and system demonstration.

EDUCATION

Diploma in ICT : Application Development

Durban University of Technology

Jan 2024 –

Nov 2026

National Matric Certificate

J.G Zuma High School

Jan 2017 –

Nov 2021

SKILLS

Languages and Frameworks

- C#
- ASP.NET MVC
- ASP.NET Core MVC
- HTML
- CSS
- Entity Framework Core
- Entity Framework 6

Databases

- SQL Server
- Azure Cosmos DB
- Azure Blob Storage

Tools and Methodologies

- Git and GitHub
- Technical Documentation
- Use Cases
- MVC Architecture
- Microsoft Azure
- Azure Blob Storage
- OAuth 2.0
- ASP.NET Identity
- ASP.NET Core Identity

CERTIFICATES

FNB Academy 2025

Full Stack Development

Introduction to Cybersecurity

Cisco Networking Academy

PROJECTS

Student Management System

Technologies used: C#, ASP.NET Core MVC, .NET 8, Microsoft Azure, Azure Cosmos DB, Azure Blob Storage, OAuth 2.0

- Built and deployed a full-stack student management portal on Microsoft Azure within a collaborative 10 person development team.
- Engineered an OAuth 2.0 authentication pipeline (GitHub and Google) paired with Role-Based Access Control (RBAC), securing administrative endpoints by enforcing a strict allow list policy.
- Implemented full CRUD operations handling multi-attribute student records, reducing administrative processing bottlenecks.
- Isolated sensitive asset storage by querying Azure Cosmos DB for structured textual data while routing profile images to Azure Blob Storage protected by time limited Shared Access Signature (SAS) tokens.

Security Management System(Pirana Guarding)

Technologies used: C#, ASP.NET MVC, SQL Server, HTML5, CSS3, JavaScript

- Developed an operational guard tracking web portal engineered to manage daily shift rotations for up to 12 active guards per site.
- Engineered an anti-fraud check-in engine using dynamically generated, daily expiring QR codes, eliminating 100% of static credential and code-reuse attendance tampering.
- Built automated business rule validation that evaluates guard shift statuses in real time, immediately blocking unauthorized login attempts for off-duty personnel.
- Designed a centralized Director Dashboard displaying real-time attendance logs, active timestamps, and incident reporting metrics.

EcoRewards SA – Smart Recycling Rewards Web Application

Technologies used: C#, ASP.NET MVC, .NET Framework 4.8, Entity Framework 6, SQL Server, Bootstrap 5, Chart.js

- Architected a Code-First relational database schema in SQL Server via Entity Framework 6 to manage user profiles, waste categories, point balances, and reward items.
- Built role-based access control via ASP.NET Identity across 3 discrete user roles (Resident, Municipal Staff, Administrator) with tailored permissions and views.
- Programmed automated reporting engines using iTextSharp and EPPlus, allowing administrators to generate and export custom monthly/annual PDF and Excel data summaries.
- Integrated interactive Chart.js dashboards to track monthly recycling trends, collection event metrics, and reward redemptions.
- Pitched and demonstrated the live software prototype to industry judges and stakeholders at Demo Day.

HerCycle, Digital Marketplace for Women Entrepreneurs in Menstrual Wellness.

Built for the Imbokodo Hackathon 2026, August 7 to 8

Technologies used: ASP.NET Core MVC, SQL Server, Entity Framework Core, ASP.NET Core Identity, Bootstrap 5, Chart.js

- Developed and presented a functional e-commerce marketplace during a 24-hour hackathon sprint, transitioning micro-entrepreneurs from informal WhatsApp sales to a centralized web platform.
- Engineered an end-to-end seller management suite featuring real-time inventory tracking, order status workflows, and interactive sales analytics.
- Built customer discovery workflows including multi-category filtering, shopping cart state management, wishlists, and checkout pipelines.
- Implemented a dual-tier subscription architecture using ASP.NET Core Identity, enforcing listing limits between Free (5 items) and Premium (50 items at R99/month) accounts.

VisionScribe – AI-Powered Image Analysis Application

Technologies used: C#, ASP.NET Core MVC, .NET 8, Azure Computer Vision API, Azure Blob Storage, IMemoryCache, REST APIs

- Architected a full-stack image intelligence application utilizing ASP.NET Core MVC (.NET 8) to parse, tag, and synthesize written descriptions from uploaded photos within seconds.
- Integrated the Azure Computer Vision REST API with Azure Blob Storage pipelines, transforming raw vision payloads (captions, dominant colors, detected objects) into structured user insights.
- Engineered an in-memory caching layer using IMemoryCache for repeated uploads, eliminating 100% of duplicate Azure Cognitive API calls and significantly cutting network latency.
- Applied SOLID principles and Dependency Injection by decoupling IBlobStorageService and IVisionAnalysisService interfaces from controllers to ensure testability and modularity.